Intercepts

1

a

i $x = 0$ ii $y = 0$

b

i $x = 3$ ii $y = 4$

c

i $x = 0$ ii $y = 0$

d

i $x = 4.5$ ii -3.5

2

a $(0, -5)$

b $(0, 3)$

3

a $y = 3$

b The y-intercept is the constant without x in the equation.

c $y = c$

4

a The constant term of 4 is the same.

b All the graphs intersect the y-axis at the point $(0, 4)$.

c The line will pass through $(0, 4)$.

5

a $y = 3$ **b** $y = -5$ **c** $y = -3$ **d** $y = 4$

e $y = 0$ **f** $y = 2$ **g** $y = \frac{2}{3}$ **h** $y = \frac{8}{5}$

6

a Yes **b** Yes **c** No **d** No

e Yes **f** No **g** Yes **h** Yes

7

a $x = 1$ **b** $x = -1$ **c** $x = 2$ **d** $x = -3$

e $x = 0$ **f** $x = 2$ **g** $x = \frac{1}{3}$ **h** $x = 2$

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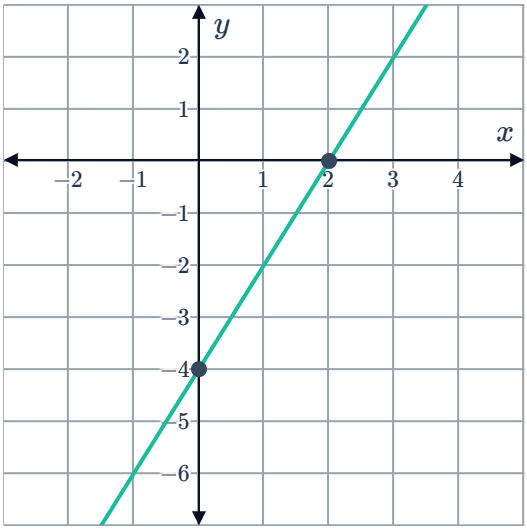
8

a

i $(0, -4)$

ii $(2, 0)$

iii

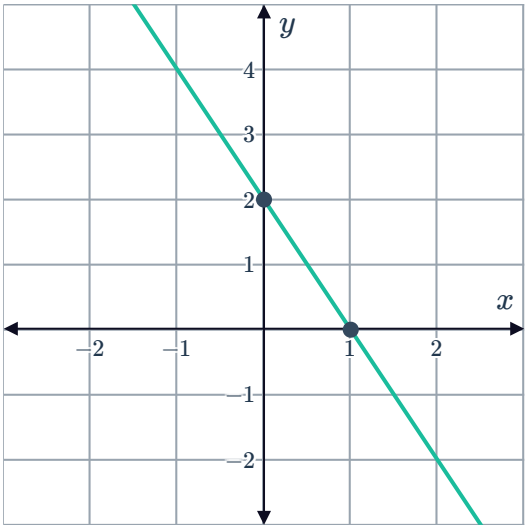


b

i $(0, 2)$

ii $(1, 0)$

iii



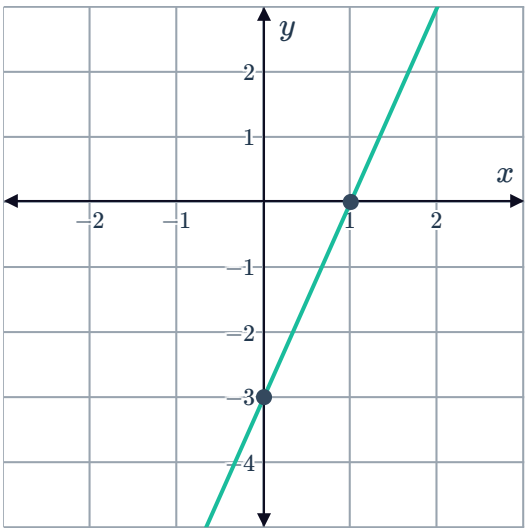
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c

i $(0, -3)$

ii $(1, 0)$

iii

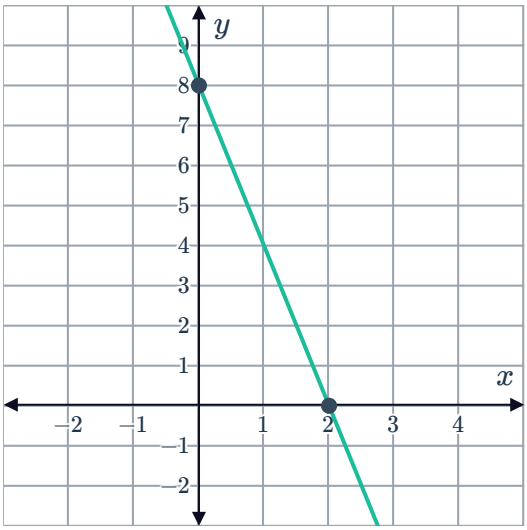


d

i $(0, 8)$

ii $(2, 0)$

iii



Lines through the origin

9

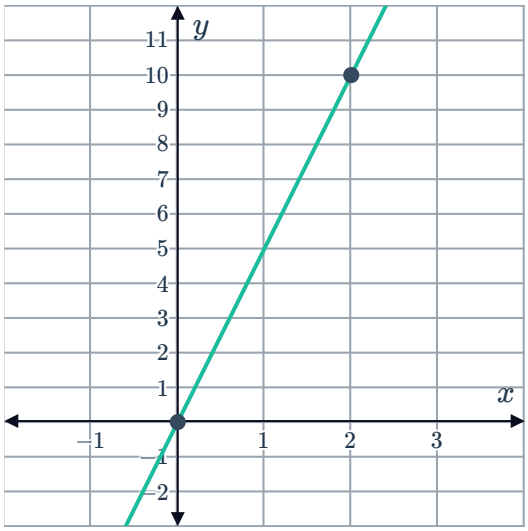
a $(0, 0)$

b $(0, 0)$

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c $y = 10$

d

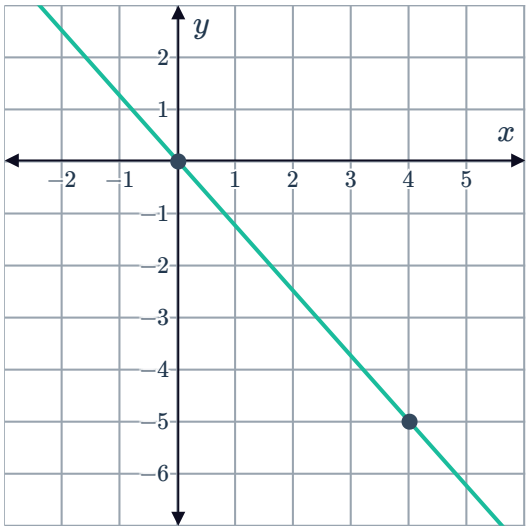


10

a $(0, 0)$

b $(4, -5)$

c



11 There is no constant term as the value of c is zero.

12

a No **b** Yes **c** Yes **d** Yes

e Yes **f** Yes **g** Yes **h** No

Gradient

13

a $y = 4$ **b** $y = 6$

c 2 **d** $m = 2$

e It is the coefficient (number in front) of x.

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14

a The coefficient of x is the same, 4.

b All the graphs have gradient of 4.

c The line will have gradient of 4.

15

a $m = 9$ **b** $m = -7$ **c** $m = \frac{5}{4}$ **d** $m = -1$

16 $y = 2 + 7x, y = 7x, y = 7x - 2$

17

a
i $m = 2$ **ii** $c = 9$

b
i $m = 5$ **ii** $c = -6$

c
i $m = -5$ **ii** $c = 8$

d
i $m = -4$ **ii** $c = -2$

e
i $m = 4$ **ii** $c = -\frac{1}{2}$

f
i $m = -2$ **ii** $c = -\frac{2}{3}$

g
i $m = 5$ **ii** $c = 4$

h
i $m = \frac{3}{2}$ **ii** $c = 3$

18

a $y = 2x + 5$ **b** $y = -3x + 2$ **c** $y = -2x - 1$ **d** $y = 4x$

e $y = -7$ **f** $y = 4$ **g** $y = \frac{1}{2}x - 2$ **h** $y = -\frac{3}{4}x + \frac{1}{2}$

Approximate solutions of equations

19

a $x \approx 1.2$ **b** $x \approx -6.7$

20

a $x \approx 2.3$ **b** $x \approx 2.3$ **c** $x \approx 0.3$ **d** $x \approx 0.3$

21

a $x \approx 2.2$ **b** $x \approx 2.3$ **c** $x \approx 1.5$ **d** $x \approx 0.6$

e $x \approx 2.9$ **f** $x \approx 13.3$ **g** $x \approx -0.1$ **h** $x \approx -10.5$

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